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How to conduct quali-quantitative digital methods research?

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Introduction

Digital methods has emerged as a prominent paradigm for how the social and cultural sciences can meaningfully respond to the current-day ubiquity of digital trace data. Following Rogers (2013), such methods are defined by the attempt to 'follow the medium' and repurpose digital traces (hyperlinks, hashtags, likes, and so on) for the study of collective life, including not least public debate and involvement. Arising from mainly qualitative, case study-based traditions in media studies and science & technology studies (STS), digital methods embraces the project of straddling the qualitative-quantitative divide, actively searching for new 'quali-quantitative' styles of analysis (Blok et al. 2017). However, the precise research procedures, method combinations, and validity criteria for this endeavor remain subject to debate (see, e.g., Munk 2019).

In this short how-to guide, we build on existing debates and our own experiences with collectively teaching and conducting digital methods research to suggest some practically useful and methodologically defensible answers to these questions. Our suggested approach involves the combination of three otherwise distinct method repertoires within the overarching project of digital methods: online ethnography or netnography (Kozinets 2019), exploratory or visual network analysis (Venturini et al. 2021), and computer-assisted content analysis (Carlsen & Ralund 2022). While good digital, quali-quantitative research is being done through different sets of methods, we argue that these three methods in particular provide strong 'complementarities' (Blok & Pedersen 2014) for many digital methods purposes. In particular, they facilitate richly situated forms of online grounding together with carefully delineated mappings of larger, socio-symbolic formations.

We focus in this guide on the affordances and pitfalls of Twitter data in particular. This is where much of our own research experience lies; where much digital methods innovation has happened (e.g. Marres & Gerlitz 2016); and Twitter remains one of the more accessible sources of large-scale social media data samples (e.g. Rafail 2018). We believe many of our methodological suggestions will be relevant for other platforms and approaches, even as they will have to be suitably adapted to the circumstances of these (see, e.g., Perriam et al. (2020) on the challenges of Facebook research in a 'post-API' era). We also affirm that, particularly when it comes to netnography, researchers will often want to adopt a cross-platform (Pearce et al. 2020) approach in the early, exploratory phases of inquiry. This will be important as a means of gauging the specific shape and dynamics of the Twitter debates.

Throughout this guide, we exemplify more general points by way of our own collective inquiry into the case of green NGO campaigning in Denmark during the early phases of the COVID-19 pandemic in the Spring of 2020. This was a setting in which the pandemic as a focal event, when seen from a green activist perspective, threatened to derail official commitments to climate policy reform that had been built up in prior years. Yet, COVID-19 *also* presented opportunities for NGOs to link the climate and COVID-19 risks to new effects.

For present purposes, our inquiry is suitable to illustrate some widespread use scenarios for quali-quantitative digital methods, in terms of mapping public debates.

Collecting Data: Scouting, Sampling, Querying

The first task of quali-quantitative digital methods inquiry is to adequately sample and collect data. Digital methods typically involve studying phenomena where the questions of *who* (actors) and *what* (topics) is uncertain to begin with. By implication, collecting data is an exploratory task, where one searches out different locations, actors, and themes of potential relevance to one's object of study. It is through this search process that we reach some sort of certainty as to the boundaries of our study. Our framework uses different means to demarcate the object of study, each with their complementary strengths and weaknesses. We use i) netnography's flexibility to scout different locations, ii) network snowball sampling to locate relevant actors, exploiting relational information in digital traces, iii) keyword search to locate the varieties of ways in which our thematic interests are articulated.

Online ethnographic methods – or what we call *netnography,* following Kozinets (2019) – aims to study interactions on social media and other online *data sites*. A netnographic campaign moves from an initial idea of what one wants to research to a broad investigation of relevant actors and their interaction activities (commenting, sharing, content production, etc.) as well as the images, hashtags, symbols, or memes that circulate between actors. Importantly, netnography is iterative and based on a learning-by-doing approach. While netnography can indeed integrate various forms of semi- or fully automated data collection techniques into its design - as our framework suggests to do - this is not initially at the heart of the method. Focus is rather on the effort that a researcher puts into trying to understand the case or issue that is being investigated in a "handheld" manner.

This also means that netnography is a highly flexible method that can be done comparatively across almost any kind of social media platform or online forum. Further, a key starting point for the early phase of a netnographic campaign is encapsulated in the idea of scouting. By way of focused attention and time spent reading, scouting provides the initial filtering of online information located via searches, whereby it "helps narrow down the choice set from which you will select your data sites" (Kozinets 2019: 215). The netnography thus provides the first starting points for the research process which can both inform the next stages in a sampling strategy, and provide the ground for doing analysis of both network ties and the use and sharing of words between and across actors.

Relative to Kozinets approach, we stress with others in digital methods (Caliandro 2018) that what netnography is ultimately oriented towards here is not 'online communities' as usually conceived, but rather the mapping of circulating practices and fragments of conversation that generate wider publics. This is important since it means the scouting operations just outlined start blending into a well-known style of data sampling: snowballing. The underlying question is: who *else* is participating in this public conversation, besides those I have already encountered netnographically? To search for answers, forms of network-based snowball sampling will often be appropriate – *provided* it is used alongside an emerging stabilization of the wider sampling frame. Let us explain what this means and how it works.

Sampling, in the widest sense, concerns the question of delimitation, that is, the selection of subsets of wider data-scapes to analyze (Gerlitz & Rieder 2013). While survey-based research standardly deploys so-called probability sampling techniques, quali-quantitative digital methods research must orient itself rather to non-probability approaches, such as snowball sampling (Rafail 2018). This always entails what, in classical network studies, is

known as the boundary specification problem (Laumann et al. 1989): whether based on definitional traits of the actors and their mutual positions, or on specific types of relationships among them, we need some way of specifying where the network starts and stops. Overall, this task has only become more complicated when using digital trace data.

Our framework relies on a useful distinction between 'topic-demarcated' and 'actor-demarcated' populations - while acknowledging that both are 'semi-bounded' at best when conducting Twitter-based studies (Rafail 2018). An actor-demarcated population might be German parliamentarians, South American immigrants working in Spain or, in our case inquiry, Danish environmental NGOs. Working with an actor-demarcated approach means using Twitter follower relations, retweeting, as well as @mentions or @replies in tweets as signs of how one's emerging sample might be reasonable expanded, and to check for obviously relevant actors otherwise in risk of being missed out. Such snowball-based techniques have been shown to enable a high degree of saturation, even as they are unlikely to yield 'perfect' samples. Saturation is achieved once snowballing as per above starts yielding only already-identified actors relevant to one's category.

These actor-based techniques can be combined, depending on purpose, with forms of topic-based sampling, focused on textual content in a debate under study. Here, we depend on constructing a query list, that is, a set of keywords to be used for searching. Any analysis claiming to be topic-demarcated is only as good as its query list. The big challenge when working with an Application Programming Interface (API) such as Twitter's is that one cannot explore the entire corpus before choosing the final query list. Rather, researchers have to choose a query set to begin with. Researchers often choose a single hashtag or a few of the most prominent words, yet this has been shown to yield quite problematic results (Rafail 2018). The problem with simply choosing the terms that seem relevant is that we seldom know precisely the words actors use to talk about a given theme before our research begins, hence finding and delimiting a set of query terms is part of the analysis.

The first step involves getting an initial sample to search from in order to expand one's query set. There might be some words we already know are relevant for our query, which we call a keyword sample set; moreover, we have a keyword scout set, which comprise the parts of the discursive landscape where we will search for new keywords to add to our sample set. In our own case inquiry, we know that 'corona' and 'covid' are relevant, and we use this to look for other covid-related terms in the wider Danish political discourse, focusing on institutional politic (using 'dkpol' as scout term) and civil society action (using 'dkcivil' as scout term). The logic is that there is a known overlap between our sample query and our scout query, as well as the possibility that relevant but hitherto unknown terms show up in the scout query. This selection of scout keywords is crucial for the inquiry, as it will be defining for the expansion of the keyword sample set. This becomes less of a problem the more reason you have to believe that your search terms will be articulated in the same way across different contexts. Conversely, it is a bigger problem the more reason you have to believe they will be articulated differently across contexts. Consulting one's netnography is useful for making informed decisions about which parts of the discursive landscape to scout.

To investigate how Danish environmental activists and NGOs were responding to the COVID-19 pandemic situation, we began our netnographic campaign by searching Twitter, and to some degree Facebook and Instagram, with a few central keywords that we intuitively thought would be fruitful starting points. Among these were #klima ('climate'), #covid19, #corona, and #krise ('crisis'). From these initial clues, we started scouting which actors used these keywords and hashtags, as well as which other words, images, videos, and memes they used. In line with this research agenda, we compiled an actor-list that resulted from this open-ended sampling process, and which contained among others a set of youth-based

activist groups and initiatives (e.g., Fridays for Future, Extinction Rebellion). We also compiled a list of the most interesting terms used alongside or in conjunction with our initial search terms. Some of these proved to be fruitful starting points for deeper investigation. Most notably, the Danish hashtag #grøngenstart ('green restart') became a significant phrase for the wider climate debate in the initial stages of the corona pandemic.

In a set of parallel explorations, we searched for more systematic ways in which to demarcate our sample population. Network-based actor sampling from Twitter yielded mostly a familiar set of environmental and other NGOs, which we further curated by way of official lists showing NGOs recognized in parliamentary policy hearing processes on climate. For purposes of focused analysis later on, this meant dropping the youth-based activist groups not found on these lists - even as we kept them for wider contextual, netnographic analysis. This actor-based demarcation had the corollary of making whatever the listed environmental NGOs were tweeting about in the relevant period - that is, their full tweet lists during Spring 2020 - our source of semantic variation. To supplement this, and be able to ground NGO discourses in the wider corona debate during this period, we constructed a secondary, topic-demarcated dataset around covid-19, using the query expansion technique detailed earlier. With these data collection steps completed, we turned our attention to data exploration.

Exploring Data: Immersion, Visualization, Word Associations

Data exploration is at the center of most quali-quantitative digital methods projects, as this is where the researcher immerses herself in specific sites or practices; tries to visualize emerging patterns of interaction; and performs the deep readings that may inform also computer-assisted content analyses of the wider corpus.

For a netnographic campaign, the key to this phase is to narrow and focus the research through *immersive engagement*. In online research, immersion means careful reading, observing, reflecting, and pondering the online interactions that one encounters on social media and elsewhere, in a way that adds depth and substance to the initial scouting done in the previous phase. Practically, immersion is facilitated through using what Kozinets (2019: 283) calls an immersion journal – a digitally-adapted version of the fieldworkers' notebook. As the inquiry progresses, this becomes a record or archive of central actors, their relations, as well as recurrent images, videos, and visual symbols deemed interesting by the researcher. One can concretely record notes in a manual notebook, or use notebook software that is shareable if working in teams (e.g., Evernote, OneNote). One can also use spreadsheet files (as we did).

An immersion journal has three interconnected purposes in the exploration phase. First, it provides a structured landscape for data exploration in which the researcher can traverse the types of online traces that have been collected, i.e., links, quotes, images, posts, etc., and look for patterns and variations across these. Second, it is useful for keeping track of key examples and cases related to the overall research theme that could be worth exploring further through in-depth immersion, whereby the investigative process is iteratively extended. Third, through detecting patterns and variations, the immersion journal provides starting points for deciding on directions to go for further research via either manual, semi-automated, or automated content and network exploration and analysis. Exploring the data in a distinct *netnographic* manner (while building bridges to the other methods), could be to take words from the immersion journal and visualizing them in word clouds (for content exploration); or to draw simple, preliminary manual networks of the relations between the most central actors that appear in one's journal.

Indeed, these explorations of the immersion journal can be fruitfully extended in terms of actor relations by drawing on so-called visual network analysis (VNA). VNA has emerged from within the digital methods paradigm as a distinct, exploratory approach to the wider field of network analysis. This approach utilizes long-standing traditions for drawing networks as point-and-line graphs, going back to Jacob Moreno's sociograms of the 1930s (Venturini et al. 2021). With digital methods, the approach takes on a topological or 'map-like' ambition, often using particular layout algorithms to create visually meaningful depictions of medium- to large-scale social media datasets. As such, the resulting graphs become tangible depictions of the polarizations and clusters making up landscapes of public debate.

Broadly speaking, visualization practices play a dual role in quali-quantitative digital methods: first, as a tool of iterative exploration (this section); and second, as a means of stabilizing analyses of key socio-symbolic patterns (next section) (Decoupere 2020). Practically, the open source Gephi software affords both steps in a user-friendly way, including for medium- to large-size datasets. As a tool of exploration, VNA helps to form hypotheses concerning interesting patterns in data not immediately obvious from the immersion. This is done, for instance, by exploring actor relations by way of follower-following, retweets, or @mentions, to indicate established social sub-clusters or ascriptions of influence and authority in debates, respectively. Here, actors are visualized as nodes that are positioned on a graph according to their connectivity (usually by way of a so-called 'force-directed layout' like ForceAtlas2 - see Venturini et al. 2021). Moreover, one can size the nodes according to their importance, such as their in-degree of @mentions, and color them according to their category, such as types of organization or similar. Altogether, these visual exploration practices provide a mapping of the entire debate space, as well as allowing for zooming in on particular sub-regions of the graph for further inspection. This latter practice, in turn, will often provide clues for new netnographic immersion.

Similarly, the netnographic exploration of a public debate can be extended and corroborated using computer-assisted content analysis (CACA). Where netnography typically has extensive contextual knowledge of a limited set of actors and their talk, CACA allows the researchers to explore large corpuses, yet with limits to the context knowledge surrounding each expression. Drawing on the netnography, CACA starts from thematic interests and related query terms, and it then expands these by searching for similar words in a large text corpus (e.g. of tweets). In order to create a space of similarity between words, we use a simple word embedding technique called word2vec. Essentially, word2vec uses neural network techniques to group together words used in similar ways (Evans & Aceves 2016). By exploring the resultant word similarity space, we develop our query set further and also start to develop potential subthemes and discover new relevant themes. We then have a list of query terms we can use to return documents to read in-depth. And we can change and update our query to visit new places in the discursive landscape, in order to saturate our understanding of our themes and to search particularly fruitful or puzzling parts of the discourse.

Our own netnographic exploration of green NGOs during COVID-19 began by traversing what we had stored in our immersion journal. Concretely, this journal for us took the shape of a spreadsheet organized according to actors (the NGOs) and their posts, images, links, as well as tentative analytical reflections about what each NGO were up to. From early on, we chose to deep dive into a few specific debates and hashtags that we followed intensely, based on their seeming centrality to the debates. One key example was the hashtag and slogan '#grøngenstart' ('green re-start'). This hashtag was evidently being circulated in the early phase of the pandemic by the Danish climate think tank CONCITO, although the genealogy of it turned out to date a bit further back (Isfeldt et al. 2022). Consequently, we

began systematically noting which actors shared this hashtag, which other words and hashtags accompanied it, and on which platforms it was most visible.

Overall, some NGOs were more active in these symbolic framing activities than others, and organizations made slightly different symbolic investments into the circulation of specific slogans, visual styles, and so on. To gain a better sense of inter-activist relations, in terms of who were endorsing which organizations' campaigning styles, we explored the retweet networks among our listed NGOs. This yielded a rather clear picture of salient sub-clusters, with a few organizations - notably the Danish 350.org (*Klimabevægelsen*) and Danish Greenpeace - acting as bridges across the NGO landscape, both retweeting and being retweeted more frequently than others. We found this interesting, in part because we knew from the netnographic immersion that Greenpeace had opted for its own symbolic tactic, sloganned as #nejtilnormal ('no to normal'), as distinct from the 'green re-start' campaign. The retweet network helped us make sense of which and how activist slogans were being dispersed.

However, we also wanted to know how far such framing efforts were being spread beyond green NGOs. To explore the articulation of green concerns in the wider COVID-19 debate on Danish Twitter, we searched for climate and biodiversity words in the expanded corpus. We continuously sampled tweets for qualitative reading, which we coded with a focus on specific climate concerns, as well as with a focus on how links were made to the corona debate. This phase of reading and coding was open, in the sense that we did not begin with a focus on on any one specific climate concern, nor with any one specific discursive link between COVID-19 and green concerns. Rather we were interested in mapping the full variations in green concerns and their link to COVID-19 - at least in this initial phase of exploration. In our exploration of climate-related terms, starting from 'climate' yielded further terms such as 'climate crisis' and 'handlekraft' (*action capacity*). We then explored in depth how the climate crisis and the corona crisis were co-articulated in the wider Twitter public. Various interesting aspects surfaced, such as: the tension between "young people's" crisis and "older people's" crisis; COVID-19 demonstrating the lack of political will regarding the climate crisis; discussion around the (in-)adequacy of addressing climate issues in the midst of a pandemic; and corona crisis response measures as a possible model for dealing with the climate crisis.

Analyzing Data: Cultural Logics, Actor Relations, Semantic Patterns

In the final, analytical phase of quali-quantitative digital methods inquiry, the goal is to reach a stabilized, well-corroborated, and persuasive account of the key cultural logics, actor relations, and semantic patterns at work in the public debate as a so-called socio-symbolic formation (Fuhse et al. 2020). Further, and importantly for the mixed-methods aspiration of our framework, a key question for validity concerns the way in which findings are mutually attuned or calibrated across the different method registers (Isfeldt et al. 2022). Sometimes, this means triangulating claims among the various methods, while at other times it is rather a matter of using each methods' strength to ground or contextualize the others.

Netnographic accounts are essentially oriented to the distilling of core cultural logics that help make sense of the wider public debate-space, or which sheds light on particularly interesting sub-themes. The road from data exploration to the analysis of cultural logics can follow a relatively classic approach in qualitative analysis. This involves preparing and collating the data for analysis, open and/or closed coding of the symbolic content, combining codes into higher order themes, as well as charting and visualizing these through outputs (e.g., word lists, word clouds, manual networks, etc.). In a crude sense, cultural logics refers

to the shared frameworks of meaning that enable individual actors to make the world sensible and communicable (Cohen 1985: 82). In the context of online interactions and digital methods, understanding cultural logics implies analyzing the ways in which actors' symbolic practices converge or diverge in recognizable patterns, in terms of their taking positions on various themes in public space.

In addition, a netnographic campaign in the context of a quali-quantitative approach as we advocate here *also* pays attention to the emerging findings of the network and content analyses. Carefully considering findings such as keyword co-occurrence across different actors, for instance, can raise new questions regarding varying semantic meaning of words and images (Krieg et al. 2017). This is also where our netnographic approach diverges to some extent from more classic forms of virtual or digital ethnography that mainly seek to adopt classic ethnographic methods to online contexts (e.g., Hine 2015, Pink et al. 2015). As a digital method, our ambition here is to make netnographic use of the affordances (tools, data, etc.) that digital environments and online public spaces provide (Caliandro 2018), in relation to the broader network and content exploration. In this respect, a key analytical move that a netnography contributes vis-a-vis these two other methods is what Kozinets (2019) refers to as telescopic and microscopic movements. The aim of such movements is to contextualize the role of individual actors, or objects that circulate between actors (e.g., images, hashtags, etc), as well as relevant subtopics, by zooming in and out between micro- and macro-perspectives, thereby explicating the cultural logics at work across different scales and levels.

Visual network analysis (VNA) provides a suitable means of further stabilizing and validating the analysis of key and recurrent interaction patterns in the public debate-space, especially as concerns actor-relational patterns in an actor-bounded Twitter population (Rafael 2018). Here, once visualized via a force-directed layout, the overall shape of the retweet or other Twitter-based networks provide a velter of information on the unfolding of debate (Venturini et al. 2021). First, the graph indicates the existence or not of more-or-less well-demarcated sub-clusters, meaning sub-groups of actors who tend to retweet each other more regularly then the rest. Second, especially in the case of middle- to large-size graphs, the polarization of the graph in terms of its 'north-south' and 'east-west' dimensions can be interpreted to suggest key social and symbolic axes of variation, difference, or dispute. Third, the graph lends itself to interpretations of network centrality as a measure of influence or authority, as well as to interpretations of actors working to bridge information flows across sub-clusters. As for these latter attributes, as well as the clustering logic of the graph, visual interpretation can be supported by use of simple network-mathematical measures.

Finally, the goal of computer-assisted content analysis (CACA) in the analytical phase is two-fold. First, to arrive at a comprehensive qualitative analysis of a given theme or category - what is its defining features, subcategories, and relations to the process under study. Second, to build a so-called classifier and analyze the distribution of the theme across central points of variation in the dataset (time, actors types, network position). By implication, the reading of text moves from being more open-ended to being more focused at addressing the theoretical interests and interpretative uncertainties related to a given theme. It is through analytical reading, combined with the network analysis and netnography, that the researcher eventually decides on what quantitative text analysis is the most interesting to pursue. Importantly, close reading provides the researcher with a grounded understanding of the theme to be quantified. Still, the qualitative reading typically yields also certain aspects of the semantic patterns that are hard to quantify - such as the relation to other themes, the style of communication, and so on. Throughout, it should be kept in mind that there are many ways in which one can quantitatively classify text as being about a certain theme

(Evans & Aceves 2016). For our approach, we recommend the use of a so-called dictionary method, where a set of words are seen as indicative of a certain theme and tweets are classified to themes if it uses a word from the dictionary. The dictionaries themselves are developed through word2vec-based search for similar words, as described above. In other words, we are building domain- or corpus-specific dictionaries.

We return now to our own inquiry and the case of the 'green re-start' hashtag. By netnographically following how it was used and who shared it, and by following how individual actors used it (the micro perspective) compared to how the hashtag was used across all actors (the macro perspective), we began noticing how it operated through particular cultural logics. It became clear that green re-start carried a message that the pandemic could be seen as an opportunity to reset the workings of society - as a kind of shock therapy - away from fossil fuel dependency towards a green transition. In this respect, the activist hashtag campaign semantically conveyed a sense of hope, and framed a horizon for a brighter future beyond the pandemic. This is likely one of the reasons why it quickly became popular among many youth-based green organizations (see figure 1).

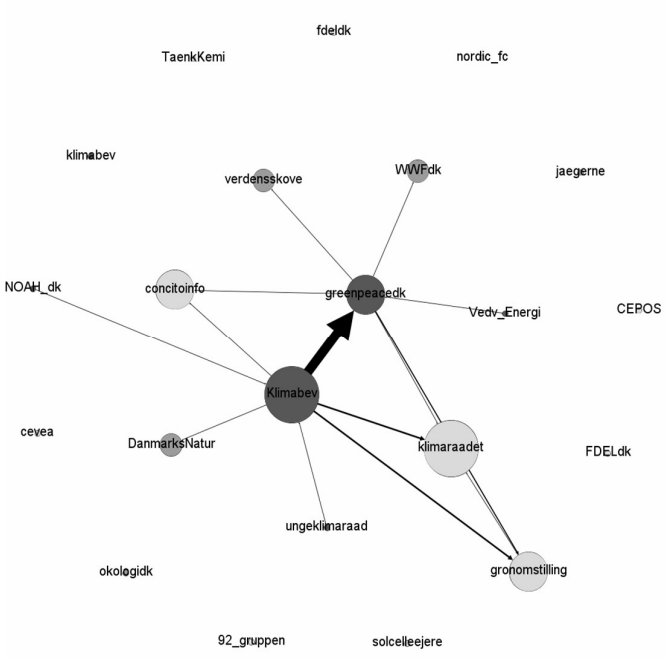
Figure 1: Deployment of 'Grøn Genstart' ('green re-start') slogan by youth-based activist group. Source: Den Grønne Studenterbevægelse (www.dgsb.dk)

However, a few months into the pandemic, after the hashtag garnered considerable traction and support from a variety of NGOs, it soon also attracted the attention of government ministries, agencies, and politicians on Twitter (Isfeldt et al. 2022). Suddenly, the hashtag - or slogan if you will - was being appropriated by actors such as the Minister for Climate and Energy, who used it openly to signal his intentions for boosting the green agenda in the midst of the pandemic. As this appropriation of the slogan by government came into effect,

the initial sense of hope that surrounded green re-start soon became supplanted - in the eyes of many NGOs and youth-based green groups - by a sense of annoyance and hopelessness, in light of their disappointment with the government's climate and environmental policies. We found additional hashtags and memes that circulated on Twitter and other platforms to display similar forms of dynamic changes in the course of the first stage of the pandemic. Yet, the particular cultural logic of the green re-start hashtag made it an interesting case of a so-called "inter-risk analysis", i.e., how green NGOs saw the corona crisis as a political opportunity to rethink the climate crisis (Enggaard et al. 2022).

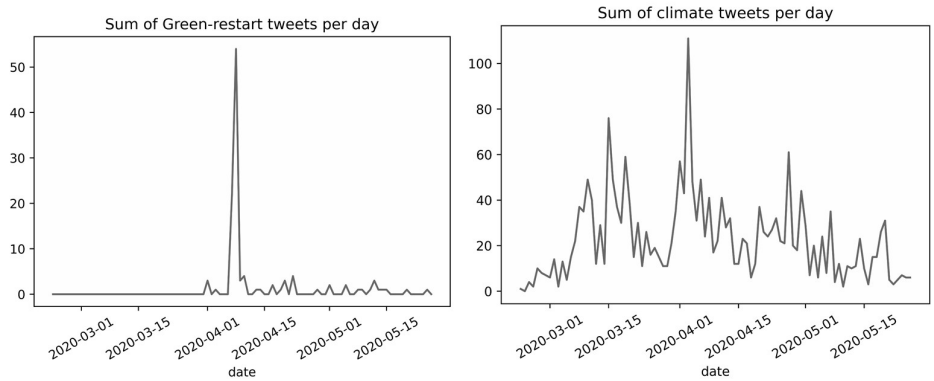
To corroborate further our sense of how diffusion of activist frames was happening among the Danish green NGOs and what this might entail for the wider debate, we categorized the NGOs into three main types (plus a residual) for purposes of visual network analysis (VNA). Hence, on figure 2 below, dark-grey nodes refer to distinctly climate-oriented NGOs; grey nodes refer to distinctly nature- and biodiversity-oriented NGOs; light-grey refers to climate-related think tanks and expert groups; and white is a residual of other organizational types (consumer protection, electric vehicle support group, and other). The categorization confirmed our initial exploration, in so far as it clearly shows the centrality for information diffusion, and thus the visibility and authority accorded to, two climate-oriented NGOs (*Klimabevægelsen* and Greenpeace). It also reveals, however, that these two NGOs often retweet the content of central think tanks and expert bodies - as indeed happened when CONCITO, a green think tank, coined the green re-start slogan. Finally, we were initially surprised to learn that nature- and biodiversity-oriented NGOs found themselves in a clearly secondary layer of inter-NGO visibility, given that NGOs in other national settings (such as Norway) seemed to more actively relate COVID-19 to a wider sense of crisis in nature (Enggaard et al. 2022). Based on our netnography, we were able to interpret this as signaling the clear predominance of distinctly climate-related campaigning in Denmark.

Figure 2: Retweet network among Danish green NGOs, Spring 2020



Finally, we hypothesized that a consequence of the green re-start campaign could be that it resulted in a shift in inter-risk attention dynamics, such that climate issues became easier to co-articulate with the COVID-19 issue. Figure 3 (left) below, based on a simple Twitter count, shows a clear sign of an intense but short-lived green-restart campaign. While the slogan itself was short-lived, it might still have resulted in a rise in climate tweets within the corona debate more generally. In order to address this question, we rely on a custom-built climate dictionary, made from searching for similar words and learning from the netnography and qualitative analysis. We ended with a list of 14 words, which we took to be indicative of the climate topic. Figure 3 (right) below, "sum of climate tweets per day", shows that while 'green re-start' did result in an episodic rise, there is no clear pattern to indicate that it shifted the discourse more generally. Hence, it seems unlikely that the co-articulation of climate and COVID-19 on Twitter in general became more prominent after the campaign.

Figure 3: The appearance of 'green re-restart' in the wider corona debate on Twitter



Reading across the methods, we draw the following interim conclusion: 'green re-start' emerged as a prominent activist intervention during the early stages of the pandemic in Denmark, as promoted by a green think-tank and youth-based activist groups. The slogan and campaign was not unanimously shared, however, among Danish green activists, with some high-profile climate NGOs (notably Greenpeace) opting for a different framing. Moreover, while causing considerable but short-lived attention on Twitter, there is no indication that the campaign left a lasting mark on wider corona debates. In our own research work, these quali-quantitative insights came to form the backbone of two extensive analyses, documenting the potentials of this digital methods approach to contribute in a timely manner to important public issues (Isfeldt et al. 2022; Enggaard et al. 2022).

Conclusion

Quali-quantitative digital methods research is on the ascendency in the socio-cultural sciences, with many novel studies and methodological contributions being added each year to what is now a sizable body of knowledge and know-how. In this how-to guide, we have emphasized a particular set of digital methods and their complementarities for many research purposes: online ethnography or netnography, exploratory or visual network analysis, and computer-assisted content analysis. When used together, we suggest, this suite of methods facilitate a combination of richly situated cultural logics with carefully demarcated mappings of larger socio-symbolic formations, typically in the shape of public

debates. As such, they afford much of the weaving together of qualitative and quantitative registers of inquiry, styles of analysis, and validity criteria that digital methods have always been after (see Blok et al. 2018; Munk 2019; Isfeldt et al. 2022).

We hope that our combination of general suggestions with a case example pertaining to our own work on green NGO campaigning during pandemic times will provide readers with sufficient inspiration to commence their own quali-quantitative digital methods inquiries. In this process, readers will likely need to consult other sources, many of which are found in our list of references, for more extended technical and methodological unfoldings of specific points or steps. Moreover, readers should take heed of what we stressed already in the introduction: while our specific approach builds on much literature and experience, there are other viable approaches to digital methods out there, qualitative, quantitative, and otherwise. We can only invite further exploration and debate on the relative merits of such diverse approaches when it comes to furthering the project of digital methods - which, after all, is fast becoming a sizable part of socio-cultural research writ large.

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